

AgriSmart AI — Required Model Report

PRIMARY FIELD RESULT Macro-F1 0.5782 Accuracy 0.5938 · Top-3 0.8828 128 held-out PlantDoc-style field images · never used for training	MODEL ImageNet-pretrained ResNet18 160 px · crop/leaf ensemble + mirror TTA 18 classes · 6 crops
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1. Dataset & split

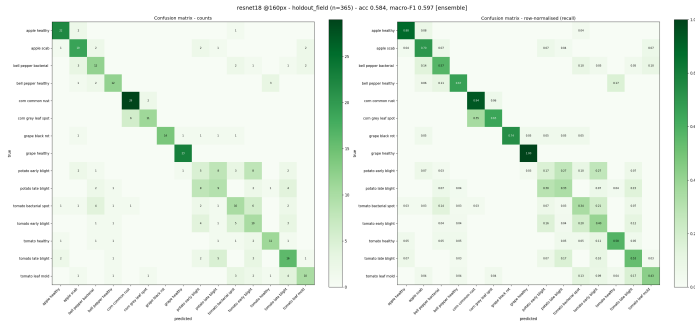
Train/validation: PlantVillage 18-class subset, seed-42 split — **6,951 / 984 images**. **Primary test:** PlantDoc-style real field photos, **128 images**, held out from training. Secondary hold-out: 365 images. Three classes absent in the primary field split are excluded from macro-F1 and disclosed in the repository.

2. Model / approach

ResNet18 fine-tuned with AdamW (lr 1.5e-4, weight decay 1e-4), cosine schedule, label smoothing 0.05, weighted sampling, resized crop, flip, rotation, colour jitter and blur. Field adaptation mixes PlantDoc with PlantVillage replay. A MobileNetV3-small + logistic-regression leaf gate rejects non-leaf photos instead of inventing a disease.

3. Metric & result

Evaluation	N	Accuracy	Macro-F1	Top-3
Held-out field — primary	128	0.5938	0.5782	0.8828
Held-out field — secondary	365	0.5836	0.5966	0.8329
Lab validation — reference	984	0.9939	0.9944	1.0000



Confusion matrix: row-normalised and count panels from the committed field evaluation artifact ('report/confusion_matrix.png').

Per-class precision / recall / F1: Potato early blight 0.250/0.250/0.250; tomato bacterial spot 0.273/0.333/0.300; potato late blight 0.333/0.375/0.353; tomato leaf mold 0.500/0.333/0.400; bell pepper bacterial spot 0.444/0.444/0.444; tomato early blight 0.400/0.667/0.500; tomato healthy 0.750/0.375/0.500; bell pepper healthy 0.667/0.500/0.571; corn grey leaf spot 0.667/0.500/0.571; apple scab 0.667/0.600/0.632; tomato late blight 0.750/0.600/0.667; apple healthy 0.778/0.778/0.778; corn common rust 0.692/0.900/0.783; grape black rot 1.000/0.750/0.857; grape healthy 0.923/1.000/0.960.

4. Baseline & safeguards

Lab-only baseline on the same field split: **accuracy 0.3047 / macro-F1 0.2986**. Shipped model: 0.5938 / 0.5782 — **+0.2796 macro-F1 (~94% relative)**. Leaf gate: ROC-AUC 0.998; refuses 96.9% of non-leaf photos and wrongly refuses ~1.1% of real leaves.

5. Limitations

Large lab→field gap (≈0.99 vs ≈0.58 macro-F1); blur, clutter, occlusion and lighting remain difficult. Scope is limited to 18 classes and six crops. Field imbalance makes low-support per-class values indicative. The gate can over-refuse. Guidance is decision support, not agronomic diagnosis; weather features need internet and degrade to manual inputs offline.